IMCS,
University of Sindh

Programming AI

BS(AI) Part-2- 2nd Semester

Prof. Dr. Imtiaz Ali Korejo

Python Data Types

Week-2 , Lecture-4

Outline of Today's Lesson

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Section-1

Python Programming

PYTHON DATA TYPES

By: Dr. Imtiaz Ali Korejo

Python Data Types

- In programming, data type is an important concept.
- Variables can store data of different types, and different types can do different things.
- Python has the following data types built-in by default, in these categories:

Text Type:	str
Numeric Types:	int, float, complex
Sequence Types:	list, tuple, range
Mapping Type:	dict
Set Types:	set, frozenset
Boolean Type:	bool
Binary Types:	bytes, bytearray, memoryview
None Type:	NoneType

Getting the Data Type

- You can get the data type of any object by using the type() function:

```
x = 5
print(type(x))
```

```
Output:
<class 'int'>
```

Setting the Data Type

- In Python, the data type is set when you assign a value to a variable:

`x = "Hello World"`	str	
`x = 20`	int	
`x = 20.5`	float	
`x = 1j`	complex	
`x = ["apple", "banana", "cherry"]`	list	
`x = ("apple", "banana", "cherry")`	tuple	
`x = range(6)`	range	
`x = {"name" : "John", "age" : 36}`	dict	
`x = {"apple", "banana", "cherry"}`	set	
`x = frozenset({"apple", "banana", "cherry"})`	frozenset	
`x = True`	bool	
`x = b"Hello"`	bytes	
`x = bytearray(5)`	bytearray	
`x = memoryview(bytes(5))`	memoryview	
`x = None`	NoneType	

Setting the Specific Data Type

- If you want to specify the data type, you can use the following constructor functions:

`x = str("Hello World")`	str
`x = int(20)`	int
`x = float(20.5)`	float
`x = complex(1j)`	complex
`x = list(("apple", "banana", "cherry"))`	list
`x = tuple(("apple", "banana", "cherry"))`	tuple
`x = range(6)`	range
`x = dict(name="John", age=36)`	dict
`x = set(("apple", "banana", "cherry"))`	set
`x = frozenset(("apple", "banana", "cherry"))`	frozenset
`x = bool(5)`	bool
`x = bytes(5)`	bytes
`x = bytearray(5)`	bytearray
`x = memoryview(bytes(5))`	memoryview

Python Numbers

- There are three numeric types in Python:
  - int
  - float
  - complex
- Variables of numeric types are created when you assign a value to them:

```
x = 1     # int
y = 2.8   # float
z = 1j    # complex
```


Complex Numbers

- Complex numbers are written with a "j" as the imaginary part:

```
x = 3+5j
y = 5j
z = -5j

print(type(x))
print(type(y))
print(type(z))
```

```
Output:
<class 'complex'>
<class 'complex'>
<class 'complex'>
```

Random Number

- Python has a built-in module called random that can be used to make random numbers:

```
import random

print(random.randrange(1, 10))
```

Python Casting

Integers:

```
x = int(1)   # x will be 1
y = int(2.8) # y will be 2
z = int("3") # z will be 3
```

Floats:

```
x = float(1)     # x will be 1.0
y = float(2.8)   # y will be 2.8
z = float("3")   # z will be 3.0
w = float("4.2") # w will be 4.2
```

Strings:

```
x = str("s1") # x will be 's1'
y = str(2)    # y will be '2'
z = str(3.0)  # z will be '3.0'
```

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Section-2

Python Programming

PYTHON STRING

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Python Strings

- Strings in python are surrounded by either single quotation marks, or double quotation marks.

`'hello'` is the same as `"hello"`.

- You can display a string literal with the print() function:

```
print("Hello")
print('Hello')
```

Quotes Inside Quotes

- You can use quotes inside a string, as long as they don't match the quotes surrounding the string:

```
print("It's alright")
print("He is called 'Johnny'")
print('He is called "Johnny"')
```

Strings are immutable sequence data type, i.e each time one makes any changes to a string, completely new string object is created

```
a_str = 'Hello World'
print(a_str)    #output will be whole string. Hello World
print(a_str[0])    #output will be first character. H
print(a_str[0:5])    #output will be first five characters. Hello
```


Assign String to a Variable

- Assigning a string to a variable is done with the variable name followed by an equal sign and the string:

```
a = "Hello"
print(a)
```


Multiline Strings

- You can assign a multiline string to a variable by using three quotes:

```
a = """Lorem ipsum dolor sit amet,
consectetur adipiscing elit,
sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua."""

print(a)
```

Strings are Arrays

- Like many other popular programming languages, strings in Python are arrays of bytes representing unicode characters.
- However, Python does not have a character data type, a single character is simply a string with a length of 1.
- Square brackets can be used to access elements of the string.

```
a = "Hello, World!"
print(a[1])
```

Get the character at index 1 (remember that the first character has the index 0):

Looping Through a String

- Since strings are arrays, we can loop through the characters in a string, with a for loop.

```
for x in "banana":
  print(x)
```

```
Output:
b
a
n
a
n
a
```

String Length

- To get the length of a string, use the len() function.

```
a = "Hello, World!"
print(len(a))
```

```
Output:
13
```

Check String

- To check if a certain phrase or character is present in a string, we can use the keyword in.

Check if "free" is present in the following text:

```
txt = "The best things in life are free!"
print("free" in txt)
```

```
Output:
True
```

Use it in an if statement:

```
txt = "The best things in life are free!"
if "free" in txt:
  print("Yes, 'free' is present.")
```

```
Output:
Yes, 'free' is present.
```

Check if NOT

- To check if a certain phrase or character is NOT present in a string, we can use the keyword not in.

Check if "expensive" is NOT present in the following text:

```
txt = "The best things in life are free!"
print("expensive" not in txt)
```

```
Output:
True
```

Use it in an if statement:

```
txt = "The best things in life are free!"
if "expensive" not in txt:
  print("No, 'expensive' is NOT present.")
```

```
Output:
No, 'expensive' is NOT present.
```

LEARNING RESOURCES

Source Reference used in Slide

- https://www.w3schools.com/python

Other Resources

- https://realpython.com/
- https://docs.python.org/3/
- https://docs.python.org/3/tutorial/index.html
- https://www.tutorialspoint.com/python/index.htm

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END of SESSION

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